(19) Write a brief description of the meaning of the notation $\lim_{x\to 8} f(x) = 25$.

(20) If $f(2) = 4$, can you conclude anything about the limit of $f(x)$ as x approaches 2?

(21) If the limit of $f(x)$ as x approaches 2 is 4, can you conclude anything about $f(2)$? Explain reasoning.

(22) Evaluate : $\lim_{x\to 3} \frac{x^2-9}{x-3}$ if it exists by finding $f(3^-)$ and $f(3^+)$.

(23) Verify the existence of $\lim_{x\to 1} f(x)$, where $f(x) = \begin{cases} \frac{|x-1|}{x-1}, & \text{for } x \neq 1 \\ 0, & \text{for } x = 1 \end{cases}$.

9.2.3 Theorems on limits

The intention of the informal discussion in the earlier section was to have an intuitive grasp of existence or non existence of the limit. However, it is neither desirable nor practical in every instance, to reach a conclusion about the existence of a limit based on a graph or table of functional values. We must be able to evaluate a limit, or discern its non existence, in a somewhat mechanical fashion. The theorems that we shall consider in this section establish such a means. The proofs of these theorems are more of technical and are beyond the scope of this textbook.

In Illustration 9.1, we concluded that $\lim_{x\to 2}(x^2+3) = 2^2+3 = 7$. That is, the limit of $f(x) = x^2 + 3$ as x tends to 2 is equal to $f(x)$ evaluated at $x = 2$. [That is, $f(2)$]. However, this process of evaluation, as noted earlier, will not always work because $f(x)$ may not even be defined at x_0. Nevertheless, it is true that **if f is a polynomial, then it is always possible to calculate the limit by evaluation**.

Theorem 9.1

Let $P(x) = a_0 + a_1x + a_2x^2 + \cdots + a_n x^n$ be a polynomial, where $a_0, a_1, \cdots, a_n$ are real numbers and n is a fixed positive integer. Then

$$\lim_{x\to x_0} P(x) = a_0 + a_1x_0 + a_2x_0^2 + \cdots + a_nx_0^n = P(x_0).$$

Example 9.7

Calculate $\lim_{x\to 3}(x^3 - 2x + 6)$.

Solution

$P(x) = x^3 - 2x + 6$ is a polynomial.

Hence, $\lim_{x\to 3} P(x) = P(3) = 3^3 - 2\times 3 + 6 = 27$.

Example 9.8

Calculate $\lim_{x\to x_0}(5)$ for any real number x_0.

Solution

$f(x) = 5$ is a polynomial (of degree 0).

Hence $\lim_{x\to x_0}(5) = f(x_0) = 5$.

The limit of a constant function is that constant.

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